

Counting and Recurrence relation

Counting is a fundamental aspect of combinatorics and discrete mathematics. It involves determining the number of ways certain events can occur.

Sum Rule:

If there are m tasks can be done in $n_1, n_2, n_3, \dots, n_m$ ways respectively with no two task can be done simultaneously then the number of ways to do one of these m tasks is $n_1 + n_2 + \dots + n_m$ ways.

Problem: A dealership has 10 sedans and 7 SUVs available. If a customer wants to buy either a sedan or an SUV, how many choices does the customer have?

Solution:

Sedans: 10 options

SUVs: 7 options

Total options: $10 + 7 = 17$

Product Rule

If a procedure is carried out by performing the tasks $t_1, t_2, \dots, t_m$. If task t_i can be done in n_i ways after the tasks $t_1, t_2, \dots, t_{i-1}$ have been done then there are $n_1 \cdot n_2 \cdot \dots \cdot n_m$ ways to carry out the procedure.

Example:

You are making a sandwich. You have 3 types of bread (white, wheat, rye) and 4 types of fillings (turkey, ham, chicken, veggie). How many different sandwiches can you make if you choose one bread and one filling?

Solution:

Bread options: 3

Filling options: 4

Total sandwiches: $3 \times 4 = 12$

Example:

How many different license plates are available if each plates contain a sequence of 3 letters followed by 4 digits?

Solution:

There are 26 choices for each of the 3 letters and 10 choices for each of the 4 digits.

Number of different license plates = $26 \times 26 \times 26 \times 10 \times 10 \times 10 \times 10$

$= 175760000$.

Permutation

A **permutation** refers to the number of ways to **arrange** a set of objects in a specific **order**, where all objects are **distinct** and **order matters**.

- **Total permutations of n distinct objects:** The number of ways to arrange n distinct objects in sequence is $n!$
- **Permutations of r objects chosen from n distinct objects:** The number of ways to arrange r objects out of n distinct objects is:

$$P(n,r) = nPr = \frac{n!}{(n-r)!}$$

Combination: An unordered selection of r elements of a set containing n distinct elements is called an r- combination of n elements and is denoted by $c(n,r)$ or nCr .

$$C(n,r) = nCr = \frac{n!}{r!(n-r)!}$$

1. Obtain the number of permutations of all the letters of the words
(i) ACCOUNTANT (ii) ENGINEERING (iii) COMMITTEE.

Solution:

- (i) The total number of letter in the word ACCOUNTANT is 10 of which 2 are A's 2 are C's 2 are T's and 2 are N's.

$$\text{Therefore, no of permutation} = \frac{10!}{2!2!2!2!}$$

- (ii) The total number of letters in the word ENGINEERING is 11 of which 3 are E's, 3 are N's, 2 are I's and 2 are G's.

$$\text{Therefore, no of permutations} = \frac{11!}{3!3!2!2!2!}$$

- (iii) The total number of letters in the word COMMITTEE is 9 of which 2 are M's, 2 are T's, and 2 are E's.

$$\text{Number of permutations} = \frac{9!}{2!2!2!}$$

2. How many words can be formed from the letters of the word 'MONDAY' if
(a) 4 letters are used at a time (b) all letters are used together and (c) all letters are used but first is a vowel.

Solution:

There are 6 letters in MONDAY

$$(i) P(6,4) = \frac{6!}{(6-4)!} = \frac{720}{(2)!} = 360$$

$$(ii) P(6,6) = \frac{6!}{(0)!} = 720$$

(iii) Since O and A are the vowels in the word MONDAY and in its permutations first place will be filled in only two ways and the rest five places may be filled by 5 ways.

therefore, the total no of words = $2! P(5,5) = 2! 5! = 240$

3. Five balls are to be placed in 3 boxes. Each can hold all the 5 balls. In how many

different ways can we place the balls so that no box is left empty , if

(a) balls and boxes are different?

(b) Balls are identical and boxes are different?

Solution:

(a) 5 balls can be distributed such that the first, second and third boxes contain 1,1,3 balls respectively.

$$\therefore \text{No of ways of distributing in this manner} = \frac{5!}{1!1!3!} = 20$$

Similarly the boxes I, II, III may contain 1, 3 and 1 balls respectively or 3,1 and 1 balls respectively. (since boxes are different).

No of ways of distributing in each of these mannners =20

Again the boxes I, II, III may contain 1, 2, 2 balls respectively or 2, 1, 2 balls respectively or 2, 2, 1 balls respectively.

$$\text{No of ways of distributing in this manner} = \frac{5!}{1!2!2!} = 30$$

$$\therefore \text{Total no of required ways } 20+20+20+30+30+30=150$$

(b) total no of ways of distributing r identical balls in n different boxes is the same as

the number of r combinations of n items, repetitions allowed.

$$C(n+r-1,r) = C(3+2-1,2)=6$$

Since 3 balls must be first put one in each of 3 boxes and the remaining 2 balls must be distributed in 3 boxes.

4. How many different words are there in the word MATHEMATICS?

Solution:

$$\text{Number of different words} = \frac{11!}{2!2!2!1!1!1!1!1!1!} = 49,89,600.$$

5. A box contains 6 white ball and 5 red balls. Find the number of ways 4 balls can be drawn from the box if a) They can be any colour
b) 2 must be red and 2 white
c) All must be of the same colour.

Solution:a) The four balls (of any colour) can be chosen from the eleven balls in

$$C(11,4) = \frac{11 \times 10 \times 9 \times 8}{1 \times 2 \times 3 \times 4} = 330 \text{ ways}$$

b)The two white balls can be chosen in 6C_2 ways and two red balls can be chosen in 5C_2 ways.

There fore the total number of ways to select 2 white balls and 2 red balls = ${}^6C_2 \times {}^5C_2$

$$= 150 \text{ ways}$$

c) Number of ways to select all 4 balls are white = 6C_4

Number of ways to select all 4 balls are red = 5C_4

Total number of ways to select fours balls are of same colour = ${}^6C_4 + {}^5C_4$
= $15 + 5 = 20$ ways

Binomial theorem

It is possible to expand $(a + b)^n$ into a sum of the form,

$$(a+b)^n = {}^nC_0 a^n b^0 + {}^nC_1 a^{n-1} b^1 + {}^nC_2 a^{n-2} b^2 + \dots + {}^nC_{n-1} a^1 b^{n-1} + {}^nC_n a^0 b^n$$

$$(a + b)^n = \sum_{r=0}^n {}^nC_r a^{n-r} b^r$$

where, $n \geq 0$ is an integer and each nC_r is a positive integer known as a binomial coefficient. The Binomial coefficients are the r- combinations from the set of n elements.

The above expansion is called the “Binomial Theorem”.

1. Find the expansion of $(x+y)^5$

Solution:

$$(x+y)^5 = C(5,0) x^5 + C(5,1)x^4 y + C(5,2)x^3 y^2 + C(5,3)x^2 y^3 + C(5,4)xy^4 + C(5,5)y^5$$

$$= x^5 + 5x^4 y + 10x^3 y^2 + 10x^2 y^3 + 5xy^4 + y^5$$

2. What is the coefficient f $x^{10}y^{15}$ in the expansion of $(x+y)^{25}$?

Solution:

From the binomial theorem $(a + b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r$

Let $a = x$, $b = y$ and $n = 25$

$$(x + y)^{25} = \sum_{r=0}^{25} \binom{25}{r} x^{25-r} y^r$$

The coefficient of $x^{10}y^{15}$ is $C(25,15) = \frac{25!}{15!10!}$

3. What is the coefficient of $x^{10}y^{15}$ in the expansion of $(3x-2y)^{25}$?

Solution:

From the binomial theorem $(a + b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r$

Let $a = 3x$, $b = -2y$ and $n = 25$

$$(3x - 2y)^{25} = \sum_{r=0}^{25} \binom{25}{r} (3x)^{25-r} (-2y)^r$$

The coefficient of $x^{10}y^{15}$ is $C(25,15) = -\frac{25!}{15!10!} 3^{10} 2^{15}$.

PIGEONHOLE PRINCIPLE

If n pigeons are accommodated in m pigeon – holes and $n > m$ then at least One pigeon hole will contain two or more pigeon. Equally if n objects are put in m boxes and $n > m$, then at least one box will contain two or more objects.

Generalization of the pigeonhole principle:

If n pigeons are accommodated in m pigeon holes and $n > m$, then one of the pigeon holes must contain at least $\left\lfloor \frac{n-1}{m} \right\rfloor + 1$ pigeons, where $\lfloor x \rfloor$ denotes the greatest integers less than or equal to x , which is a real number.

1. Among 100 people, show that atleast 9 of them were born in the same month?

Solution:

Here $n=100$ and $m=12$ months.

Then $\left\lfloor \frac{100-1}{12} \right\rfloor + 1 = 9$ were born in the same month.

2. Show that if seven colors are used to paint 50 bicycles, atleast 8 bicycles will be the same color.

Solution:

Here $n=50$, and $m=7$.

By Extended pigeonhole principle, $\lfloor \frac{50-1}{7} \rfloor + 1 = 8$

Therefore at least 8 bicycles will have the same color.

RECURRENCE RELATION:

A recurrence relation for the sequence for the sequence $\{a_n\}$ is an equation that expresses " a_n " in terms of the previous terms of the sequence, namely $a_0, a_1, a_2, a_3, \dots, a_{n-1}$, for all integers n with $n \geq n_0$, where n_0 is a non negative integer.

Example 1:

The **recurrence relation for the Tower of Hanoi** problem is given by:

$F(n) = 2F(n-1) + 1$ for $n > 0$, where $T(0) = 0$.

The solution to this recurrence relation is **$F(n) = 2^n - 1$** .

This means that the minimum number of moves required to solve the Tower of Hanoi puzzle with n disks is **$2^n - 1$** .

Example 2:

Write down the recurrence relation for the Fibonacci numbers.

Solution:

The **recurrence relation for the Fibonacci numbers** is given by:

$F(n) = F(n-1) + F(n-2)$ for $n \geq 3$, where $F(1) = 1$ and $F(2) = 1$.

The above will give the Fibonacci sequence $\{1, 1, 2, 3, 5, 8, 13, 21, \dots\}$.

Example 3:

The sequence is given by $f(n)=3 \cdot 2^n, n \geq 0$. Find the recurrence relation.

Solution:

For $n=0$, $f(0) = 3 \cdot 2^0 = 3$.

For $n \geq 1$, $f(n)=3 \cdot 2^n$

$$f(n-1) = 3 \cdot 2^{(n-1)} = \frac{3 \cdot 2^n}{2} = f(n)/2$$

Therefore $f(n) = 2 \cdot f(n-1)$ for $n \geq 1$ with $f(0) = 3$.

Example 4:

Find the recurrence relation for the sequence $f(n)=2n+9, n \geq 0$.

Solution:

For $n=0$, $f(0) = 0+9 = 9$.

For $n \geq 1$, $f(n)=2n+9$

$$f(n-1) = 2(n-1)+9 = 2n-2+9 = (2n+9) - 2 = f(n)-2$$

Therefore $f(n) = f(n-1)+2$ for $n \geq 1$ with $f(0) = 9$.

Example 4:

Find the recurrence relation for the sequence $f(n)=2n^2+1, n \geq 0$.

Solution:

For $n=0$, $f(0)=2 \cdot 0+1 = 1$

For $n=1$, $f(1)=2 \cdot 1 + 1 = 3$

$$\text{For } n \geq 2, f(n) = 2n^2 + 1 \quad \dots (1)$$

$$f(n-1) = 2(n-1)^2 + 1 = 2(n^2 - 2n + 1) + 1 \quad \dots (2)$$

$$f(n-2) = 2(n-2)^2 + 1 = 2(n^2 - 4n + 4) + 1 \quad \dots (3)$$

subtracting (2) from (1)

$$f(n) - f(n-1) = (2n^2 + 1) - (2(n^2 - 2n + 1) + 1)$$

$$= 4n - 2 \quad \dots (4)$$

$$f(n-1) - f(n-2) = (2(n^2 - 2n + 1) + 1) - (2(n^2 - 4n + 4) + 1)$$

$$= 4n - 6 \quad \dots (5)$$

Subtracting (5) from (4)

$$f(n) - f(n-1) - (f(n-1) - f(n-2)) = 4n - 2 - 4n + 6$$

$$f(n) - 2f(n-1) + f(n-2) = 4, n \geq 2 \text{ with } f(0) = 1, f(1) = 3.$$